

Agent: human

Which buildings in Exeter have green roofs?

Agent: host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to find buildings in Exeter with green roofs. Provide the general steps to solve this geospatial query.'}

Agent: planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is buildings with green roofs, and we need to find as many entries as possible that meet this condition.

output steps: ["Find Exeter 1 area", "Search for buildings in Exeter with green roofs as many search results"]

Agent: host_agent

Tool send_message returned result : The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is buildings with green roofs, and we need to find as many entries as possible that meet this condition.

output steps: ["Find Exeter 1 area", "Search for buildings in Exeter with green roofs as many search results"]

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

Agent: named_area

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the

area polygon for Exeter..,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

Agent: data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons that represent Exeter using multiple relevant columns
    # We look for 'Exeter' in name1_text and also check for relevant description/descriptiongroup
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter', na=False))) &
                  (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement', 'City', 'Town', 'Suburban Area']))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement-related description.'
    return [summary, artifact_name, artifact_description, filtered]

filter_exeter_area_polygon(data)
```

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons that represent Exeter using multiple relevant columns
    # We look for 'Exeter' in name1_text and also check for relevant description/descriptiongroup
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter', na=False))) &
                  (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement', 'City', 'Town', 'Suburban Area']))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement-related description.'
    return [summary, artifact_name, artifact_description, filtered]
```

```
# Call the function with the provided data list
filter_exeter_area_polygon([exeter_area_polygon])
```

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons that represent Exeter using multiple relevant columns
    # We look for 'Exeter' in name1_text and also check for relevant description/descriptiongroup
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter', na=False))) &
                  (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement', 'City', 'Town', 'Suburban Area']))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement-related description.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered to 1 polygon(s) for Exeter using name and description columns.',]

Output from agent data_analysis_agent : Filtered to 1 polygon for Exeter using name and description columns. Artifact name: filtered_exeter_area_polygon
Description: GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement-related description. If you need to view or analyze this polygon further, let me know!

Agent: named_area

I found the area polygon for Exeter. The filtered artifact is named **filtered_exeter_area_polygon**. It contains only the polygon(s) representing Exeter, filtered by name and settlement-related description.

Agent: host_agent

Calling tool send_message with args : {'target': 'buildings', 'task_description': "Search for buildings with green roofs within the area defined by the artifact 'filtered_exeter_area_polygon'."}

Agent: buildings

Calling tool call_os_ngd with args : {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'all_buildings_exeter_green_roof_search'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are :
['buildings_all_buildings_exeter_green_roof_search',
 'buildingline_all_buildings_exeter_green_roof_search',
 'buildingpart_all_buildings_exeter_green_roof_search'],\n Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for buildings with green roofs within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Search for buildings with green roofs within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Search for buildings with green roofs within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None."],\n counts of records fetched are :
[69169, 69342, 85480].\n ', [, ,]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'buildings_all_buildings_exeter_green_roof_search' to find buildings with green roofs. Save the filtered results as 'buildings_exeter_green_roofs_only'. Consider roof color or material attributes that indicate a green roof."}

Agent: data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['buildings_all_buildings_exeter_green_roof_search']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_green_roofs(data: list):
    import geopandas as gpd
    df = data[0]
    # Filter for green roofs using both the green roof presence and primary material columns
    green_roof_mask = (
        (df['roofmaterial_greenroofpresence'].str.lower() == 'present') |
        (df['roofmaterial_primarymaterial'].str.lower() == 'green roof')
    )
    filtered = df[green_roof_mask].copy()
    summary = f"Filtered buildings with green roofs: {len(filtered)} entries found."
```

```

    artifact_name = 'buildings_exeter_green_roofs_only'
    artifact_description = (
        "A GeoDataFrame containing only buildings in Exeter identified as having green roofs. "
        "Filtering is based on either 'roofmaterial_greenroofpresence' being 'Present' or "
        "'roofmaterial_primarymaterial' being 'Green Roof'. All relevant columns are retained."
    )
    return [summary, artifact_name, artifact_description, filtered]

filter_green_roofs(data)

```

Tool Code Executor

```

import matplotlib
matplotlib.use('Agg')
def filter_green_roofs(data: list):
    import geopandas as gpd
    df = data[0]
    # Filter for green roofs using both the green roof presence and primary material columns
    green_roof_mask = (
        (df['roofmaterial_greenroofpresence'].str.lower() == 'present') |
        (df['roofmaterial_primarymaterial'].str.lower() == 'green roof')
    )
    filtered = df[green_roof_mask].copy()
    summary = f"Filtered buildings with green roofs: {len(filtered)} entries found."
    artifact_name = 'buildings_exeter_green_roofs_only'
    artifact_description = (
        "A GeoDataFrame containing only buildings in Exeter identified as having green roofs. "
        "Filtering is based on either 'roofmaterial_greenroofpresence' being 'Present' or "
        "'roofmaterial_primarymaterial' being 'Green Roof'. All relevant columns are retained."
    )
    return [summary, artifact_name, artifact_description, filtered]

# Call the function with the provided data list
filter_green_roofs([buildings_all_buildings_exeter_green_roof_search])

```

Tool Code Executor

```

import matplotlib
matplotlib.use('Agg')
def filter_green_roofs(data: list):
    import geopandas as gpd
    df = data[0]
    # Filter for green roofs using both the green roof presence and primary material columns
    green_roof_mask = (
        (df['roofmaterial_greenroofpresence'].str.lower() == 'present') |

```

```
        (df['roofmaterial_primarymaterial'].str.lower() == 'green roof')
    )
    filtered = df[green_roof_mask].copy()
    summary = f"Filtered buildings with green roofs: {len(filtered)} entries found."
    artifact_name = 'buildings_exeter_green_roofs_only'
    artifact_description = (
        "A GeoDataFrame containing only buildings in Exeter identified as having green roofs. "
        "Filtering is based on either 'roofmaterial_greenroofpresence' being 'Present' or "
        "'roofmaterial_primarymaterial' being 'Green Roof'. All relevant columns are retained."
    )
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered buildings with green roofs: 225 entries found.',]

Output from agent data_analysis_agent : Filtered buildings with green roofs: 225 entries found. An artifact has been generated with the name 'buildings_exeter_green_roofs_only'. This artifact is a GeoDataFrame containing only buildings in Exeter identified as having green roofs. Filtering is based on either 'roofmaterial_greenroofpresence' being 'Present' or 'roofmaterial_primarymaterial' being 'Green Roof'. All relevant columns are retained.

Agent: buildings

Search Results: - 225 buildings with green roofs were found within the area defined by the artifact 'filtered_exeter_area_polygon'.

Filtered Artifact: - buildings_exeter_green_roofs_only: This artifact contains only the buildings in Exeter identified as having green roofs. The filtering was based on either the 'roofmaterial_greenroofpresence' being 'Present' or the 'roofmaterial_primarymaterial' being 'Green Roof'.

If you need further analysis or details about these buildings, please specify!

Agent: host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot the buildings with green roofs in Exeter using the artifact 'buildings_exeter_green_roofs_only' and overlay the Exeter area boundary from 'filtered_exeter_area_polygon'."}

Agent: plotting_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['buildings_exeter_green_roofs_only', 'filtered_exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def plot_green_roof_buildings_exeter(data: list):
    import geopandas as gpd
    import folium
    import pandas as pd
    from shapely.geometry import mapping

    # Unpack data
    buildings = data[0].copy()
    exeter_area = data[1].copy()

    # Remove non-serializable columns (all datetime columns) for folium plotting
    datetime_cols = buildings.select_dtypes(include=['datetime64[ns]', 'datetime64[ns, UTC]']).columns
    buildings = buildings.drop(columns=datetime_cols)

    datetime_cols_area = exeter_area.select_dtypes(include=['datetime64[ns]', 'datetime64[ns, UTC]']).columns
    exeter_area = exeter_area.drop(columns=datetime_cols_area)

    # Ensure CRS is EPSG:4326 for folium
    if buildings.crs is not None and buildings.crs.to_epsg() != 4326:
        buildings = buildings.to_crs(epsg=4326)
    if exeter_area.crs is not None and exeter_area.crs.to_epsg() != 4326:
        exeter_area = exeter_area.to_crs(epsg=4326)

    # Get centroid of Exeter area for map centering
    exeter_centroid = exeter_area.geometry.iloc[0].centroid
    m = folium.Map(location=[exeter_centroid.y, exeter_centroid.x], zoom_start=13, tiles='cartodbpositron')

    # Plot Exeter area boundary
    folium.GeoJson(
        exeter_area.geometry.iloc[0],
        name='Exeter Area',
        style_function=lambda x: {'color': 'blue', 'weight': 3, 'fill': False}
    ).add_to(m)

    # Plot green roof buildings
    folium.GeoJson(
        buildings,
        name='Green Roof Buildings',
        style_function=lambda x: {'color': 'green', 'weight': 1, 'fillColor': 'green', 'fillOpacity': 0.5}
    ).add_to(m)

    folium.LayerControl().add_to(m)

    # Save map
```

```
map_filename = 'exeter_green_roof_buildings_map.html'
m.save(map_filename)

# Prepare summary
preview = buildings[['osid', 'description', 'roofmaterial_primarymaterial', 'roofmaterial_greenroofpresence', 'geometry_area_m2']].head(5)
summary = f"Plotted {len(buildings)} buildings with green roofs in Exeter. First 5 entries:\n{preview.to_string(index=False)}"
artifact_name = 'exeter_green_roof_buildings_map'
artifact_description = 'Folium map showing all buildings with green roofs in Exeter, overlaid with the Exeter area boundary.'
return [summary, artifact_name, artifact_description, map_filename]
```

Tool code_executor returned result : ['Plotted 225 buildings with green roofs in Exeter. First 5 entries:\n osid description roofmaterial_primarymaterial roofmaterial_greenroofpresence geometry_area_m2\n561b518f-9f84-4094-af4b-29824f626a2c Detached House Mixed Present 123.368\n9735b460-c808-4584-8236-ea144b4de340 Detached House Waterproof Membrane Or Concrete Present 123.401\n57c51e3b-f663-48ba-99bc-cb053058fed5 Detached House Metal Present 178.640\n01abd25c-4f14-4eb7-93ca-bbaf5c126697 Detached House Mixed Present 123.526\n6fd7f89f-1716-4070-81b4-f23861111668 Domestic Outbuilding Green Roof Present 40.214',]

Output from agent plotting_agent : Plotted 225 buildings with green roofs in Exeter. First 5 entries: osid description roofmaterial_primarymaterial roofmaterial_greenroofpresence geometry_area_m2 561b518f-9f84-4094-af4b-29824f626a2c Detached House Mixed Present 123.368 9735b460-c808-4584-8236-ea144b4de340 Detached House Waterproof Membrane Or Concrete Present 123.401 57c51e3b-f663-48ba-99bc-cb053058fed5 Detached House Metal Present 178.640 01abd25c-4f14-4eb7-93ca-bbaf5c126697 Detached House Mixed Present 123.526 6fd7f89f-1716-4070-81b4-f23861111668 Domestic Outbuilding Green Roof Present 40.214

- Artifact name: exeter_green_roof_buildings_map
- Description: Folium map showing all buildings with green roofs in Exeter, overlaid with the Exeter area boundary.
- Map file: exeter_green_roof_buildings_map.html

Agent: host_agent

There are 225 buildings in Exeter identified as having green roofs. These were found by searching within the Exeter area boundary and filtering for buildings where a green roof is present.

- The buildings include various types, such as detached houses and domestic outbuildings.
- The results are based on attributes like 'roofmaterial_greenroofpresence' and 'roofmaterial_primarymaterial'.

A map has been generated showing all these buildings with green roofs, overlaid with the Exeter area boundary. If you would like to view the map or need more details about specific buildings, let me know!